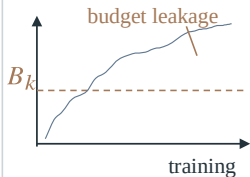


Penalty view

Costs enter the objective

Fixed penalty

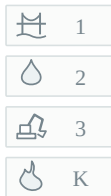


Fixed prices can leave budget leakage.
(schematic)

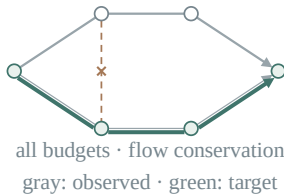
SafeOT

One flow, all budgets

Soft costs



Entropic projection



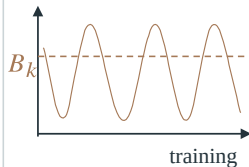
× hard constraints remove edges

Safety prices before violations

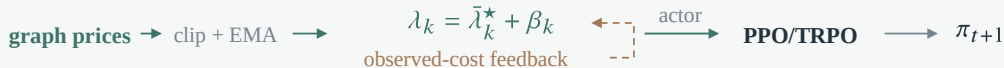
Constraint view

Budgets bound costs

Lagrangian feedback



Observed-cost feedback can lag and oscillate.
(schematic)



One price rule
Fixed penalty

$c = 0, \beta_k$ fixed

exact settings
Lagrangian

$c = 0, \beta_k$ adaptive

SafeOT

finite ε

LP

$\varepsilon \rightarrow 0$

conditional limits
Active-set

$\varepsilon \rightarrow \infty$, saturated clip